# Features of Waves

**Exam code:** 0625 & 0972

Download PDF

**Written by:** Katie M
**Reviewed by:** Caroline Carroll
Updated on 10 February 2026

Did this video help you? Yes No

## Waves & energy transfer

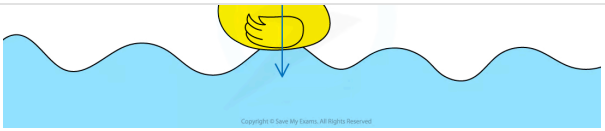
- Waves transfer **energy without** transferring **matter**
  - For sound waves, this means it is the **wave** and not the **air molecules** (the matter) itself that travels
- Objects floating on water provide evidence that waves only transfer energy and **not** matter
  - It is possible to see objects on the surface of the water bob up and down but not change their position
  - This is because the **wave** and not the **water** (the matter) itself that travels
- Waves are described as **oscillations** or **vibrations** about a fixed point
  - For example, **ripples** cause particles of water to oscillate up and down
  - **Sound** waves cause particles of air to vibrate back and forth

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

### Worked Example

The diagram below shows a toy duck bobbing up and down on top of the surface of some water, as waves pass it underneath.



Use this image

Explain how the toy duck demonstrates that waves do not transfer matter.

**Answer:**

- The plastic duck moves up and down but does not travel with the wave along the surface of the water
- The water waves transfer **energy**, but the water particles do not move
- This means when a wave travels between two points, no matter travels with it, the points on the wave vibrate up and down about **fixed** positions
- Objects floating on the water bob up and down when waves pass under them, demonstrating that there is **no movement of matter** in the direction of the wave, only **energy**

### Examiner Tips and Tricks

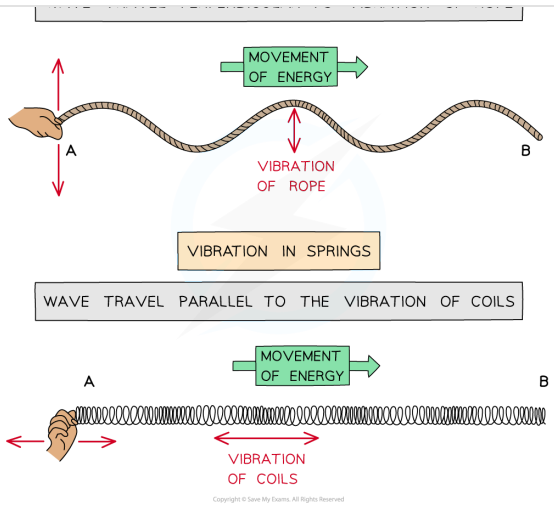
There is a key distinction between the particles (or oscillations) of a wave, and the wave itself.

The motion of the wave **causes** the particles to move. The particles themselves are not the wave.

# Wave motion

- Wave motion can be illustrated by:
  - vibrations in ropes and springs
  - experiments using water waves

## Wave vibrations

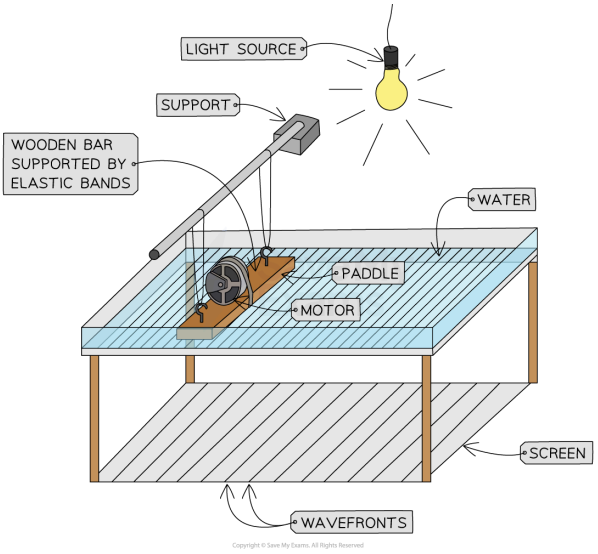



Use this image

***Waves can be shown through vibrations in ropes or springs***

- Properties of waves can be observed using water waves in a **ripple tank**
  - These properties include frequency, wavelength, amplitude, and wave speed

## Wave motion in a ripple tank




Use this image

***Wave motion of water waves may be demonstrated using a ripple tank***

# Features of a Wave

- When describing wave motion, there are several terms which are important to know, including:
  - wavefront

- crest (peak)
- trough
- wave speed

# Wavefront

- Wavefronts are a useful way of picturing waves from above
- Each wavefront, drawn as a single line, is used to represent a single wave
- The image below illustrates how wavefronts are visualised:
  - The arrow shows the direction the wave is moving and is sometimes called a **ray**
  - The space between each wavefront represents the **wavelength**
  - When the wavefronts are **close together**, this represents a wave with a **short** wavelength
  - When the wavefronts are **far apart**, this represents a wave with a **long** wavelength

## Wavefronts as viewed from above

Use this image

***Diagram showing a wave moving to the right, drawn as a series of wavefronts***

# Amplitude

- Amplitude is defined as:

**The maximum displacement of a wave from its undisturbed position**

- It is given the symbol $A$ and is measured in **metres (m)**
- On a graph where the vertical axis is **displacement**, amplitude is measured from the **undisturbed position** to either the highest point of the wave (peak) or the lowest point (trough)

# Wavelength

- Wavelength is defined as:

**The distance from one point on the wave to the same point on the next wave**

- In a **transverse** wave:

    - The wavelength can be measured from the centre of one compression to the centre of the next

- Wavelength is given the symbol $\lambda$ (lambda) and is measured in **metres (m)**
- On a graph where the horizontal axis is **distance**, the wavelength can be determined by measuring the distance from one point on the wave to the same point on the next wave

### Using a graph to determine wavelength and amplitude

Use this image

***Diagram showing the amplitude and wavelength of a wave***

## Frequency

- Frequency is defined as:

  **The number of waves passing a point in a second**

- Frequency is given the symbol $f$ and is measured in **hertz (Hz)**

## Crests & troughs

- A crest, or a peak, is defined as:

**The highest point on a wave above its undisturbed position**

- A trough is defined as

**The lowest point on a wave below its undisturbed position**

### Wave crests and troughs

## Wave speed

- Wave speed is the speed at which **energy** is transferred through a medium
- Wave speed is defined as:

**The distance travelled by a wave each second**

- The equation used to calculate wave speed is explained in The wave equation

### Worked Example

Small water waves are created in a ripple tank by a wooden bar. The wooden bar vibrates up and down hitting the surface of the water. The diagram below shows a cross-section of the ripple tank and water.

Use this image

Identify the letter which shows:

a) the amplitude of a water wave.

b) the wavelength of the water wave.

**Answer:**

Part (a)

**Step 1: Recall the definition of amplitude**

- Amplitude = The distance from the undisturbed position to the peak or trough of a wave

**Step 2: Mark the undisturbed position on the wave**

- This is the centre of the wave

Use this image

**Step 3: Identify the arrow between the undisturbed position and a peak**

**Step 1: Recall the definition of wavelength**

- Wavelength = The distance from one point on the wave to the same point on the next wave

**Step 2: Draw lines on each horizontal arrow**

- This helps to identify the points on the wave the arrows are referring to

Use this image

**Step 3: Identify the arrow between two of the same points on the wave**

- The wavelength is shown by arrow **C**

Teacher only

## New! Exclusive Teacher-Only Questions

- Fresh, expert-written questions
- Students can't access them
- Fair accurate tests - built fast

Explore in Test Builder

Join
Past Papers
Solution Banks
Sitemap

Profile
Subscription
Settings
Log out

**Company**
About us
Exam Specificity
Content Quality
Promotions
Jobs
Terms
Privacy
Cookie Policy
Help and Support

**Subjects**
Biology
Chemistry
Physics
Maths
Geography
English Literature
Psychology
All Subjects

© Copyright 2015–2026 Save My Exams Ltd. All Rights Reserved.
Neither IBO nor College Board were involved in the production of, and do not endorse, the resources created by Save My Exams.